

HOTS Question 1

In a cyber-physical robotic system, an ARM-based embedded processor must execute multiple real-time tasks including motor control, sensor fusion, and communication.

Question:

Critically analyze how memory system mechanisms (cache, memory hierarchy, MMU, and address translation) influence real-time determinism in embedded robotic applications.

In your answer:

1. Explain how cache architecture can both improve performance and reduce predictability in time-critical control systems.
 2. Evaluate the trade-offs between using physical memory access (MPU-based protection) and virtual memory (MMU-based translation) in safety-critical robotics.
 3. Discuss how address translation and cache coherence mechanisms affect worst-case execution time (WCET) estimation.
 4. Propose architectural strategies to balance high performance with deterministic behavior in embedded robotic controllers.
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HOTS Question 2

ARM Cortex processors support multiple instruction formats (ARM, Thumb-2), conditional execution, barrier instructions, and specialized data-processing operations.

Question:

Evaluate how instruction set architecture (ISA) features influence computational efficiency and reliability in embedded robotic control systems.

In your answer:

1. Compare Thumb-2 and standard ARM instruction sets in terms of code density, execution efficiency, and suitability for embedded robotics.
2. Analyze the role of conditional branching, saturation operations, and barrier instructions in ensuring numerical stability and correct synchronization.
3. Discuss how instruction-level design decisions impact power consumption and timing predictability in robotic applications.
4. Justify how hardware–software co-design principles guide the selection of ISA features for performance-critical embedded systems.